Tesamorelin

Labeled dose: 2 mg SC daily; smaller doses speculative

Tesamorelin is a stabilized growth hormone-releasing hormone (GHRH) analog that increases endogenous GH and IGF-1. [10,19] Visceral-fat reduction, liver-fat reduction, improved lipid measures, and increased lean mass is established in specific populations, and likely in others. [1–8] Muscle-composition and phosphocreatine-recovery findings support possible benefits for muscle quality and bioenergetics, while cognitive benefits remain plausible but inconsistent across trials. [12,13,17,22–24]

💉 💉 💉 **Human trials characterize common adverse effects and metabolic safety through approximately 52 weeks. Longer-term inference differs for glucose, fluid retention, and proliferative outcomes, as discussed below. [3–5,19]**

⭐ ⭐ ⭐ **Visceral-fat reduction, liver-fat reduction, improved lipid measures is established in adults with HIV-associated abdominal fat accumulation. [2–5]**

⭐ **Recovery, fitness, and generalized body recomposition through GH-axis support remain plausible applications, supported mainly by indirect or surrogate outcomes. [10,12,24,27]**

Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.

Dosage & Patterns

The pivotal phase 3 tesamorelin trials used 2 mg subcutaneously once daily for 26 weeks to reduce visceral fat in adults with HIV-associated abdominal fat accumulation. [2,4,5] A 12-week randomized, placebo-controlled trial in adults with type 2 diabetes compared tesamorelin 1 mg and 2 mg subcutaneously once daily. Both doses significantly increased IGF-1 relative to placebo, with mean increases of 33 ng/mL at 1 mg and 66 ng/mL at 2 mg. [11] Current formulations have different labeled doses: EGRIFTA WR 1.28 mg and EGRIFTA SV 1.4 mg subcutaneously once daily; they are not interchangeable dose-for-dose with each other or the original formulation. [31,32]

The evidence-based dose is

- **Tesamorelin 2 mg daily.**

Other, speculative dosing approaches commonly discussed are:

- **Tesamorelin 0.5–1 mg daily,** often selected in community practice in an effort to improve tolerability.

Reported schedules include once daily, five days per week, or every other day.

Tesamorelin

Labeled dose: 2 mg SC daily; smaller doses speculative

Trials generally used continuous daily treatment, including courses lasting 52 weeks. Clinical response, glucose status, age-adjusted IGF-1, and adverse effects inform continued treatment. [3–5,19] Persistent IGF-1 above +3 SDS warrants consideration of discontinuation; clinically important glucose deterioration or fluid-retention symptoms also require reassessment. [19] Restarting only after abnormalities have resolved or are adequately controlled is a conservative, biologically plausible approach requiring clinical reassessment.

Community protocols report 12-week ON / 6-week OFF patterns and longer courses of 12–24 weeks ON followed by 4–12 weeks OFF. These schedules are speculative and do not replace glucose and IGF-1 monitoring.

The continuous 26- and 52-week trial schedules support treatment without preset breaks when response and tolerability remain acceptable. Periodic OFF periods are a mechanistically plausible way to reduce cumulative exposure during repeated use, but do not establish prevention of delayed harms; visceral fat tends to return after withdrawal. [3–5,19]

Benefits

- **Visceral fat reduction:** Tesamorelin 2 mg subcutaneously once daily produced a pooled placebo-adjusted 15.4% reduction in visceral adipose tissue at 26 weeks. It also reduced waist circumference and waist-to-hip ratio while largely preserving abdominal subcutaneous fat. [5]
- **Improved lipid profile:** In pooled phase 3 trials, tesamorelin reduced triglycerides and improved the ratio of total cholesterol to high-density lipoprotein cholesterol. [5]
- **Reduced liver fat:** A 12-month randomized trial in adults with HIV and fatty liver found reduced liver fat. Less fibrosis progression was a secondary finding, not proof of reversal of established fibrosis. [7]
- **Plausible cognitive improvement**: A 20-week trial in healthy older adults and adults with mild cognitive impairment found improved overall cognition and executive function. [17] A later HIV trial did not establish a between-group cognitive benefit, and a small 2026 pilot found no conventional between-group cognitive improvement. [22,23]
- **Probable increased lean body mass:** In a 12-week randomized dose-ranging trial, tesamorelin 1 mg and 2 mg daily increased lean body mass relative to placebo. [1]
- **Plausible improved muscle composition:** An exploratory analysis compared 193 tesamorelin visceral-fat responders with 148 placebo recipients and found favorable changes in trunk-muscle CT density and area. [12]
- **Plausible improved fat quality:** A responder-selected exploratory CT analysis found increased visceral and subcutaneous fat density independent of changes in fat quantity, consistent with possible improvement in fat quality. [13]
- **Improved metabolic and vascular markers:** In a 12-month randomized trial in adults with abdominal obesity and reduced GH secretion, tesamorelin improved triglycerides, C-reactive protein, and carotid intima-media thickness. [8]
- **Plausible support for muscle bioenergetics:** In a 12-month substudy enrolling 39 adults with obesity and reduced GH secretion, increases in IGF-1 were associated with improved

phosphocreatine recovery after exercise. This association supports a possible bioenergetic mechanism. [24]

Applications

- **Excess visceral fat.** [2,4,5,8]
- **Fatty liver.** [7]
- **Cognition, recovery, fitness, and generalized body recomposition**. [10,17,22–24]

Cautions

- **Active malignancy:** Tesamorelin should not be used with active malignancy. [19]
- **Disrupted hypothalamic-pituitary function:** Tesamorelin is contraindicated with disruption of this axis due to hypophysectomy, hypopituitarism, pituitary tumor or surgery, head irradiation, or head trauma. [19]
- **Previous malignancy:** A preexisting malignancy should be inactive and its treatment complete before tesamorelin is considered. [19,21]
- **Pregnancy or breastfeeding**: Pregnancy is contraindicated. Product labeling advises against breastfeeding in mothers with HIV because of transmission risk; infant safety during tesamorelin exposure is not established. [19,31,32]
- **Diabetes or glucose intolerance:** Assess glucose before and during treatment. In the original trials, HbA1c reached at least 6.5% in 5% of tesamorelin versus 1% of placebo recipients by week 26. Patients with diabetes also require monitoring for retinopathy. [19,31,32]
- **Persistently elevated IGF-1:** Reassess use when insulin-like growth factor 1 remains above the age-adjusted range. Discontinuation should be considered when it remains more than three standard deviations above the age-adjusted mean. [19]
- **Concerning growth-related symptoms:** Unexplained masses, rapid tissue growth, or other concerning symptoms warrant clinical assessment and reassessment of continued use. [19,21]

Side Effects

- **Fluid retention:** Peripheral edema can occur and is often most apparent early in treatment. [19]
- **Musculoskeletal symptoms:** Joint pain and muscle pain can occur. [19]
- **Headache and nausea:** Both have been reported in clinical studies. [1–4,11]
- **Changes in sleep:** Deeper sleep and vivid dreams are reported in peptide communities and remain speculative side effects.

Cancer Risk

Tesamorelin increases circulating IGF-1 which participates in cellular proliferation. Higher circulating levels are epidemiologically associated with several cancers. It is therefore plausible that sustained growth-hormone and insulin-like-growth-factor-1 elevation could promote the growth of an occult or recurrent malignancy. [18–21]

This document provides educational summaries of limited and evolving peptide data. Nothing here is medical advice for diagnosis or treatment. © 2026 Researcher6076

Drivers of Diminished Response and Mitigation Strategies

GHRH-receptor regulation and desensitization plausibly influence response during repeated stimulation. Neutralizing antibodies were detected in vitro in some trial participants, but mean visceral-fat reduction and IGF-1 responses were similar in participants with and without them. Antibody-mediated loss of clinical response remains a plausible concern rather than an established outcome. [19] IGF-1-mediated negative feedback is established endocrine physiology. Feedback and intracellular signaling adaptation plausibly influence response, but their contribution to progressive loss of tesamorelin's clinical benefits remains uncertain. [3–5,20]

Treatment breaks are an empirical practice whose ability to preserve response remains speculative. Laboratory findings, clinical response, and adverse effects provide a basis for reassessing continued use. [3–5,19]

GH Increase Beyond a Single Administration

Although Tesamorelin's plasma half-life is approximately 26–38 minutes, its effects on endogenous GH pulsatility persist substantially longer. In a human study, Stanley and colleagues administered tesamorelin 2 mg SC daily for 14 days and measured overnight GH secretion at 10-minute intervals, beginning approximately 8 hours after the final dose. Mean overnight GH increased significantly, and mean GH pulse area increased approximately 2.5-fold, without a significant increase in pulse frequency by the primary analysis. These findings indicate that tesamorelin treatment increases the magnitude of naturally occurring GH pulses many hours after circulating drug concentrations have substantially declined. [10,19]

The biological explanation for this effect remains uncertain. Sustained intracellular signaling after GHRH-receptor activation, prolonged receptor occupancy, greater somatotroph responsiveness or releasable GH availability, and adaptation to repeated stimulation are plausible mechanisms. Interactions with endogenous GHRH, somatostatin, and GH/IGF-1 feedback may contribute. Tesamorelin probably enhances multiple successive endogenous GH pulses through effects persisting beyond peptide clearance; the repeated-dose study does not isolate a single injection from adaptations to the preceding two weeks of treatment. [10,19,20]

This effect suggests considering alternative dosing patterns.

2 mg once daily versus 1 mg twice daily

FDA clinical-pharmacology data compare 1 mg with 2 mg once daily in healthy adults, with the following findings. [28]

Measurement	*2 mg once daily versus 1 mg once daily*
GH exposure (area under effect curve)	Approximately 68% higher
Baseline-corrected IGF-1 response	Approximately 26% greater
Plasma tesamorelin accumulation	No accumulation observed during daily dosing
IGF-1 time course	Increased over days, appearing to approach a maximum around day 13

Doubling the administered dose from 1 to 2 mg therefore produced a less-than-doubling of measured GH exposure. [28]

The FDA's 1 mg versus 2 mg findings permit a useful calculation, provided the assumptions are explicit.

If two 1 mg injections produced fully recoverable independent responses, they would achieve a 19% advantage in GH output.. This conditional arithmetic is a mechanistic extrapolation of the FDA dose-response ratio; it is not measured 24-hour secretion or a validated prediction, and does not model other GH activity or feedback. [28]

The second 1 mg response would need to exceed 68% of the first to favor splitting in this simple model; at 68%, the two responses sum to the reported 2 mg response. Whether the second response reaches that threshold remains a physiological question. [28]

But it gives us a biologically informative relationship:

- If the second response is nearly as strong as the first, splitting has a plausible advantage for cumulative acute GH secretion.
- If the second response is substantially suppressed by feedback or reduced pituitary responsiveness, the advantage could disappear.

Why the longer-lasting GH effect changes this calculation

Suppose much of Tesamorelin's benefit comes from persistent changes in somatotroph function rather than the immediate GH response. In that case, we cannot assume that two administrations create two independent long-duration effects. The underlying cellular processes may already be stimulated by the first exposure.

Alternatively, a second exposure could reinforce signaling before the cellular response returns to baseline.

This distinction is particularly important because the delayed GH response is one of the most striking findings in the evidence.

This document provides educational summaries of limited and evolving peptide data. Nothing here is medical advice for diagnosis or treatment. © 2026 Researcher6076

With split dosing, a redistribution of GH secretion is more readily predicted than a substantial increase in IGF-1.

The argument for splitting is strongest if the biological effect is relatively brief and saturable. It becomes less compelling if a single exposure already induces sustained biological signaling. The observation of effects eight hours after dosing consequently weakens a simple argument that more frequent dosing must be better. It does not, however, eliminate the possible benefit of splitting, because the source of that prolonged effect is unknown.

Insulin Environment

Insulin effects depend on tissue, concentration, and metabolic context. In human hepatoma cells, insulin increased total GH-receptor abundance but exerted biphasic effects on surface receptors and GH signaling. [29] In type 1 diabetes, intraperitoneal insulin delivery increased IGF-1 and GH-binding protein, supporting a role for portal insulin in hepatic GH sensitivity. [30] Insulin effects on GH secretion must therefore be distinguished from effects on hepatic GH action. A low-insulin overnight window does not by itself predict greater IGF-1 production. Whether a fasted morning dose or a morning/bedtime split benefits from different insulin environments remains a mechanistic hypothesis.

Final mechanistic judgment

PK pattern: With the same formulation and total daily amount, splitting 2 mg into two 1 mg injections is expected to produce lower individual peaks and two exposure windows. This is a PK inference, not a clinical superiority result. [28]

GH pulse distribution: Twice-daily dosing would plausibly create a more distributed pattern of GH stimulation than once-daily dosing, although not necessarily two discrete GH pulses. (Moderate confidence)

Total daily GH secretion: Two 1 mg administrations might generate more cumulative GH than one 2 mg administration if the acute response is saturable and the pituitary responds sufficiently to the second dose. (Low confidence)

Overall assessment

For immediate summed GH responses, the saturating dose-response provides a conditional mechanistic rationale for 1 mg twice daily if the second response remains sufficiently preserved. This is a weak, speculative preference within the stated model. [28]

The prolonged GH effects after the original once-daily schedule suggest that splitting might not add independent or proportionate biological stimulation. [10]

The original 2 mg once-daily schedule remains the clinical comparator. The FDA review also describes 2 mg twice daily versus 2 mg once daily for 14 days in adults aged 70–85 years: GH and

This document provides educational summaries of limited and evolving peptide data. Nothing here is medical advice for diagnosis or treatment. © 2026 Researcher6076

IGF-1 increased more with twice-daily administration, but total daily dose doubled. That study does not establish superiority of 1 mg twice daily over 2 mg once daily. [28]

Biological Mechanism

Tesamorelin is a stabilized analog of growth-hormone-releasing hormone that activates the growth-hormone-releasing-hormone receptor on pituitary somatotrophs. These receptors couple primarily to the stimulatory G protein Gs, increasing adenylyl cyclase activity, cyclic adenosine monophosphate, and protein kinase A signaling that supports growth-hormone synthesis and secretion. [10,19,20]

Tesamorelin increases basal and pulsatile endogenous growth-hormone secretion in humans. [10]

Tesamorelin-induced GH pulses activate hepatic GH receptors and increase circulating IGF-1. Its metabolic effects plausibly involve increased lipolysis and lipid oxidation and altered very-low-density-lipoprotein metabolism. [7,10,15,20] GH-receptor signaling includes JAK2-dependent STAT and Shc/MAPK pathways, with cell-dependent STAT5 activation. [33] IGF-1 can support protein synthesis in cultured adult human muscle cells and mitochondrial remodeling in differentiating mouse myoblasts; these are mechanistic findings rather than direct tesamorelin functional outcomes. [34,35]

Tesamorelin contains an amino-terminal trans-3-hexenoyl modification that increases resistance to enzymatic degradation and improves stability relative to native growth-hormone-releasing hormone. [10,19]

For the tesamorelin acetate formulation, mean elimination half-life was 26 minutes in healthy subjects and 38 minutes in participants with HIV after 14 consecutive days of subcutaneous dosing. Peptide clearance does not define the duration of its hormonal effects: increased basal and pulsatile GH secretion was measured overnight, beginning approximately eight hours after the final dose in a two-week human study. [10,19]

References

[1] Falutz J, Allas S, Kotler D, et al. A placebo-controlled, dose-ranging study of a growth hormone releasing factor in HIV-infected patients with abdominal fat accumulation. *AIDS*. 2005;19(12):1279-1287. doi:10.1097/01.aids.0000180099.35146.30

[2] Falutz J, Allas S, Blot K, et al. Metabolic effects of a growth hormone-releasing factor in patients with HIV. *N Engl J Med*. 2007;357(23):2359-2370. doi:10.1056/nejmoa072375

[3] Falutz J, Allas S, Mamputu JC, et al. Long-term safety and effects of tesamorelin, a growth hormone-releasing factor analogue, in HIV patients with abdominal fat accumulation. *AIDS*. 2008;22(14):1719-1728. doi:10.1097/qad.0b013e32830a5058

[4] Falutz J, Potvin D, Mamputu JC, et al. Effects of tesamorelin, a growth hormone-releasing factor, in HIV-infected patients with abdominal fat accumulation: a randomized placebo-controlled trial with a safety extension. *J Acquir Immune Defic Syndr*. 2010;53(3):311-322. doi:10.1097/qai.0b013e3181cbdaff

[5] Falutz J, Mamputu JC, Potvin D, et al. Effects of tesamorelin (TH9507), a growth hormone-releasing factor analog, in human immunodeficiency virus-infected patients with excess abdominal fat: a pooled analysis of two multicenter, double-blind placebo-controlled phase 3 trials with safety extension data. *J Clin Endocrinol Metab*. 2010;95(9):4291-4304. doi:10.1210/jc.2010-0490

[6] Stanley TL, Feldpausch MN, Oh J, et al. Effect of tesamorelin on visceral fat and liver fat in HIV-infected patients with abdominal fat accumulation: a randomized clinical trial. *JAMA*. 2014;312(4):380-389. doi:10.1001/jama.2014.8334

[7] Stanley TL, Fourman LT, Feldpausch MN, et al. Effects of tesamorelin on non-alcoholic fatty liver disease in HIV: a randomised, double-blind, multicentre trial. *Lancet HIV*. 2019;6(12):e821-e830. doi:10.1016/s2352-3018(19)30338-8

[8] Makimura H, Feldpausch MN, Rope AM, et al. Metabolic effects of a growth hormone-releasing factor in obese subjects with reduced growth hormone secretion: a randomized controlled trial. *J Clin Endocrinol Metab*. 2012;97(12):4769-4779. doi:10.1210/jc.2012-2794

[9] Koutkia P, Canavan B, Breu J, Torriani M, Kissko J, Grinspoon S. Growth hormone-releasing hormone in HIV-infected men with lipodystrophy: a randomized controlled trial. *JAMA*. 2004;292(2):210-218. doi:10.1001/jama.292.2.210

[10] Stanley TL, Chen CY, Branch KL, Makimura H, Grinspoon SK. Effects of a growth hormone-releasing hormone analog on endogenous GH pulsatility and insulin sensitivity in healthy men. *J Clin Endocrinol Metab*. 2011;96(1):150-158. doi:10.1210/jc.2010-1587

[11] Clemmons DR, Miller S, Mamputu JC. Safety and metabolic effects of tesamorelin, a growth hormone-releasing factor analogue, in patients with type 2 diabetes: A randomized, placebo-controlled trial. *PLoS One*. 2017;12(6):e0179538. doi:10.1371/journal.pone.0179538

[12] Adrian S, Scherzinger A, Sanyal A, et al. The Growth Hormone Releasing Hormone Analogue, Tesamorelin, Decreases Muscle Fat and Increases Muscle Area in Adults with HIV. *J Frailty Aging*. 2019;8(3):154-159. doi:10.14283/jfa.2018.45

[13] Lake JE, La K, Erlandson KM, et al. Tesamorelin improves fat quality independent of changes in fat quantity. *AIDS*. 2021;35(9):1395-1402. doi:10.1097/qad.0000000000002897

[14] Stanley TL, Falutz J, Mamputu JC, Soulban G, Potvin D, Grinspoon SK. Effects of tesamorelin on inflammatory markers in HIV patients with excess abdominal fat: relationship with visceral adipose reduction. *AIDS*. 2011;25(10):1281-1288. doi:10.1097/qad.0b013e328347f3f1

[15] Fourman LT, Billingsley JM, Agyapong G, et al. Effects of tesamorelin on hepatic transcriptomic signatures in HIV-associated NAFLD. *JCI Insight*. 2020;5(16):e140134. doi:10.1172/jci.insight.140134

[16] Russo SC, Ockene MW, Arpante AK, et al. Efficacy and safety of tesamorelin in people with HIV on integrase inhibitors. *AIDS*. 2024;38(12):1758-1764. doi:10.1097/qad.0000000000003965

[17] Baker LD, Barsness SM, Borson S, et al. Effects of growth hormone–releasing hormone on cognitive function in adults with mild cognitive impairment and healthy older adults: results of a controlled trial. *Arch Neurol*. 2012;69(11):1420-1429. doi:10.1001/archneurol.2012.1970

[18] Qian F, Huo D. Circulating Insulin-Like Growth Factor-1 and Risk of Total and 19 Site-Specific Cancers: Cohort Study Analyses from the UK Biobank. *Cancer Epidemiol Biomarkers Prev*. 2020;29(11):2332-2342. doi:10.1158/1055-9965.epi-20-0743

[19] U.S. Food and Drug Administration. EGRIFTA (tesamorelin acetate) original prescribing information. Revised November 2010. https://www.accessdata.fda.gov/drugsatfda_docs/nda/2010/022505Orig1s000Lbl.pdf

[20] Smith RG. Development of growth hormone secretagogues. *Endocr Rev*. 2005;26(3):346-360. doi:10.1210/er.2004-0019

[21] Boguszewski MCS, Boguszewski CL, Chemaitilly W, et al. Safety of growth hormone replacement in survivors of cancer and intracranial and pituitary tumours: a consensus statement. *Eur J Endocrinol*. 2022;186(6):P35-P52. doi:10.1530/eje-21-1186

[22] Ellis RJ, Vaida F, Hu K, et al. Effects of Tesamorelin on Neurocognitive Impairment in Persons With HIV and Abdominal Obesity. *J Infect Dis*. 2025;231(5):1230-1238. doi:10.1093/infdis/jiaf012

[23] Stewart CE, French KP, Wright TJ, et al. The effect of growth hormone-releasing hormone on cognition and brain connectivity in adults with cognition ranging from normal to mild cognitive impairment. *eNeurologicalSci*. 2026;44:100616. doi:10.1016/j.ensci.2026.100616

[24] Makimura H, Murphy CA, Feldpausch MN, Grinspoon SK. The effects of tesamorelin on phosphocreatine recovery in obese subjects with reduced GH. *J Clin Endocrinol Metab*. 2014;99(1):338-343. doi:10.1210/jc.2013-3436

[25] Badran AS, Helal A, Shata KS, Ayesh H. Body composition, hepatic fat, metabolic, and safety outcomes of Tesamorelin, a GHRH analogue, in HIV-associated lipodystrophy: A meta-analysis

of randomized controlled trials. *Obes Res Clin Pract*. 2026;20(1):2-12. doi:10.1016/j.orcp.2026.01.002

[26] Ditta AM, Naeem RM, Sami MM, et al. Efficacy and Safety of Tesamorelin in People Living With HIV (PLWH) With Lipodystrophy: A Systematic Review and Meta-Analysis. *J Int Assoc Provid AIDS Care*. 2026;25:23259582261475549. doi:10.1177/23259582261475549

[27] Doessing S, Heinemeier KM, Holm L, et al. Growth hormone stimulates the collagen synthesis in human tendon and skeletal muscle without affecting myofibrillar protein synthesis. *J Physiol*. 2010;588(Pt 2):341-351. doi:10.1113/jphysiol.2009.179325

[28] U.S. Food and Drug Administration. EGRIFTA: clinical pharmacology and biopharmaceutics review, NDA 22-505. 2010. https://www.accessdata.fda.gov/drugsatfda_docs/nda/2010/022505Orig1s000ClinPharmR.pdf

[29] Leung KC, Doyle N, Ballesteros M, Waters MJ, Ho KK. Insulin regulation of human hepatic growth hormone receptors: divergent effects on biosynthesis and surface translocation. *J Clin Endocrinol Metab*. 2000;85(12):4712-4720. https://doi.org/10.1210/jcem.85.12.7017

[30] Hanaire-Broutin H, Sallerin-Caute B, Poncet MF, et al. Effect of intraperitoneal insulin delivery on growth hormone binding protein, insulin-like growth factor (IGF)-I, and IGF-binding protein-3 in IDDM. *Diabetologia*. 1996;39(12):1498-1504. https://doi.org/10.1007/s001250050604

[31] Theratechnologies Inc. EGRIFTA WR prescribing information. Revised March 2025. https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=839334d3-8c1d-4c26-9036-2ab524a6ea75

[32] Theratechnologies Inc. EGRIFTA SV prescribing information. https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=3d783378-b02d-4f19-99dd-0fc91a042224

[33] Han Y, Leaman DW, Watling D, et al. Participation of JAK and STAT proteins in growth hormone-induced signaling. *J Biol Chem*. 1996;271(10):5947-5952. doi:10.1074/jbc.271.10.5947

[34] Guan X, Yan Q, Wang D, Du G, Zhou J. IGF-1 Signaling Regulates Mitochondrial Remodeling during Myogenic Differentiation. *Nutrients*. 2022;14(6):1249. doi:10.3390/nu14061249

[35] Crown AL, He XL, Holly JM, Lightman SL, Stewart CE. Characterisation of the IGF system in a primary adult human skeletal muscle cell model, and comparison of the effects of insulin and IGF-I on protein metabolism. *J Endocrinol*. 2000;167(3):403-415. doi:10.1677/joe.0.1670403

This document provides educational summaries of limited and evolving peptide data. Nothing here is medical advice for diagnosis or treatment. © 2026 Researcher6076